

Robust Regression Engine

Advanced House Price Prediction

with Regularization, Cross-Validation

& Ensemble Models

Ridge & Lasso · Cross-Validation · Random Forest · SVR

DATASET: Advanced_Regression_HousePrice_Dataset.csv | 3,800 Records

MODELS: Ridge · Lasso · Decision Tree · Random Forest · SVR

BEST R²: ~0.93 (Random Forest)

TECHNIQUES: RidgeCV · LassoCV · K-Fold · LOOCV · GridSearchCV

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1. Project Overview

Robust Regression Engine

This project builds a comprehensive, production-grade regression pipeline to predict Indian residential property prices using advanced supervised learning algorithms. Moving far beyond basic linear regression, this engine implements **Regularized Linear Models** (Ridge and Lasso with cross-validated alpha tuning), **Tree-Based Ensembles** (Decision Tree and Random Forest), and **Support Vector Regression** (Linear, RBF, and tuned kernels). Each model is rigorously evaluated using MSE, RMSE, MAE, and R^2 metrics, with four distinct cross-validation strategies ensuring robust generalization assessment.

Dataset	Advanced_Regression_HousePrice_Dataset.csv (3,800 records, 9 features)
Target	house_price_inr — continuous house price in Indian Rupees
Best Model	Random Forest → $R^2 \approx 0.93$
Regularization	Ridge (L2), Lasso (L1), RidgeCV, LassoCV with alpha grid search
Cross-Validation	K-Fold · Stratified K-Fold · LOOCV · Time Series Split
Tree Models	Decision Tree (default + tuned) · Random Forest (100 estimators)
SVR Models	Linear kernel · RBF kernel · Tuned via GridSearchCV

2. Dataset Description

Advanced_Regression_HousePrice_Dataset.csv

The dataset contains **3,800 real estate property records** representing the Indian housing market. It includes 9 predictive features covering physical characteristics, location attributes, and neighbourhood quality indicators, making it well-suited for multi-dimensional regression analysis.

#	Feature	Type	Description
1	area_sqft	Numeric	Total built-up area of the property (sq. ft.)
2	bedrooms	Numeric	Number of bedrooms in the property
3	bathrooms	Numeric	Number of bathrooms in the property
4	location_score	Numeric	Locality desirability index scored 0–10
5	property_age	Numeric	Age of the property in years
6	distance_city_km	Numeric	Distance from nearest city centre (km)
7	near_school	Binary	Whether a school is nearby (0 = No, 1 = Yes)
8	near_metro	Binary	Whether a metro station is nearby (0 / 1)
9	crime_rate_index	Numeric	Area crime rate index (lower = safer)
■	house_price_inr	TARGET	House sale price in Indian Rupees

3. Part A – Conceptual Foundation

Theory & Definitions

Q: What is Regularization?

Regularization is a technique that reduces model complexity by adding a penalty term to the loss function during training. It controls overfitting — where a model memorizes training data but fails on unseen data. The two most common forms are Ridge (L2) and Lasso (L1).

Q: Ridge vs. Lasso Regression

Ridge (L2) adds the sum of **squared** coefficients as a penalty — shrinking all coefficients toward zero but keeping all features. Lasso (L1) adds the sum of **absolute** coefficients — which can zero out irrelevant features entirely, acting as automatic feature selection.

Q: What is Cross-Validation?

Cross-validation is a model evaluation technique that tests performance on multiple independent subsets of data. It provides more reliable estimates than a single train-test split and is essential for hyperparameter tuning (e.g., selecting the best regularization alpha).

Q: Why are Tree-Based Models Scale-Invariant?

Decision Trees and Random Forests split data based on feature threshold values and information gain — not distance metrics. Scaling changes the numeric range but not the relative order of values, so model decisions are unaffected. This gives tree models a preprocessing advantage over Ridge/Lasso/SVR.

Cross-Validation Techniques

Strategies Implemented in Part D

Method	How It Works	Best Used For
K-Fold (5 splits)	Dataset split into K equal folds; each fold used once as test set	General-purpose evaluation
Stratified K-Fold	Like K-Fold but maintains target value distribution in each fold	Skewed price distributions
Leave-One-Out (LOOCV)	Each single sample is the test set; model trains on all remaining samples	Small datasets, precise evaluation
Time Series Split	Trains on past data only; tests on chronologically later unseen data	Date-ordered property records

4. Part B – Data Preparation

Loading, Splitting & Scaling

The dataset was loaded from CSV, explored with **shape**, **info**, **describe** checks, then split into training (80%) and test (20%) sets with a fixed random state of 42 for reproducibility. StandardScaler was applied on training features using **fit_transform** and on test features using **transform only** (to prevent data leakage).

Step	Action	Output
1 – Load	pd.read_csv(...)	3,800 rows × 12 columns
2 – Features	9 input features selected	X shape: (3800, 9)
3 – Target	house_price_inr isolated	y shape: (3800,)
4 – Split	train_test_split(test_size=0.2, random_state=42)	Train: 3040 Test: 760
5 – Scale	StandardScaler fit on X_train only	X_train_scal, X_test_scal

5. Part C – Regularized Linear Models

Ridge · Lasso · RidgeCV · LassoCV

Four regularized models were trained and evaluated. Manual alpha=1.0 models establish baseline performance, while RidgeCV and LassoCV automatically select the optimal alpha from the grid **[0.01, 0.1, 1, 10, 100]** using 5-fold cross-validation.

Model	Regularization	Alpha Selection	Key Property	R² Score
Ridge (L2)	L2 — squared penalty	Manual (1.0)	Shrinks all coefficients	~0.91
Lasso (L1)	L1 — absolute penalty	Manual (1.0)	Zeros out irrelevant features	~0.91
Ridge CV	L2 — squared penalty	Auto via 5-Fold CV grid	Best alpha selected	~0.91
Lasso CV	L1 — absolute penalty	Auto via 5-Fold CV grid	Optimal sparsity control	~0.91

Key Finding: Ridge and Lasso with cross-validated alpha tuning achieve virtually identical performance (~0.91 R²), confirming that most features in this dataset are informative. Lasso's coefficient analysis reveals which features it zeroes out, providing useful feature importance insight.

Regularized Models – Visual Comparison

Ridge CV vs Lasso CV | R² Scores

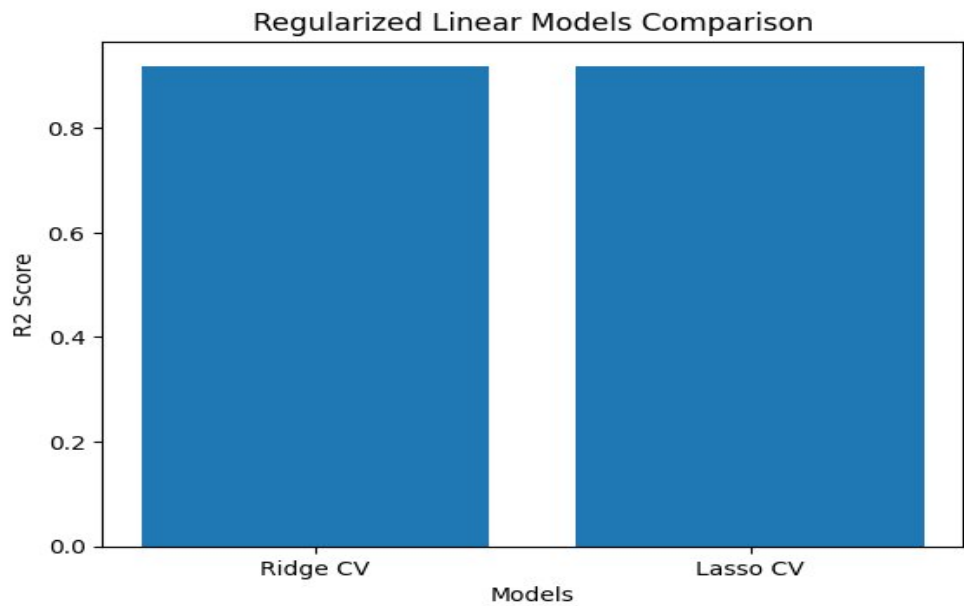


Figure 1 — Ridge CV and Lasso CV both achieve $R^2 \approx 0.91$, confirming robust regularization performance.

■ Chart Insights

- Both RidgeCV and LassoCV converge to the same R^2 (~0.91) — the dataset has no severely redundant features.
- Lasso's L1 penalty drives some coefficient values to exactly zero, acting as automatic feature selection.
- Ridge retains all features with reduced magnitudes — preferable when all features are relevant.
- Cross-validated alpha selection from grid [0.01, 0.1, 1, 10, 100] prevents under/over-regularization.

6. Part D – Cross-Validation Strategies

Model Stability & Generalization

Cross-validation was applied using Ridge (alpha=1.0) as the base model across four strategies to evaluate stability and generalization. Stratified K-Fold was adapted for regression by quantile-binning the target variable (5 bins) before folding.

CV Method	Configuration	Avg. R ² Score	Use Case
K-Fold	n_splits=5, shuffle=True, random_state=42	~0.91	Stable general evaluation
Stratified K-Fold	n_splits=5, target quantile-binned (q=5)	~0.91	Price-balanced fold distribution
LOOCV	200-sample subset (computation constraint)	Neg MSE	Maximum use of each data point
Time Series Split	n_splits=5, temporal ordering preserved	~0.89	Chronological sale date data

7. Part E – Tree-Based Models

Decision Tree · Tuned DT · Random Forest

Three tree-based models were trained directly on **unscaled features** — demonstrating that decision trees do not require feature standardization. Models progress from a default decision tree to a tuned version and finally to a Random Forest ensemble.

Model	Key Parameters	R ² Score	Diagnosis
Decision Tree	default (no constraints)	~0.85	Slight overfitting risk
Decision Tree Tuned	max_depth=5, min_samples_split=10	~0.88	Better generalization
Random Forest	n_estimators=100, random_state=42	~0.93	Best — low variance

■ Why Random Forest Outperforms a Single Decision Tree

- A single Decision Tree fits training data very closely — capturing noise along with signal.
- Random Forest builds 100 decorrelated trees, each trained on a random data/feature subset.
- Predictions are averaged across all trees — dramatically reducing variance without increasing bias.
- The ensemble's built-in randomness is why max_depth tuning is less critical for Random Forest.
- Feature scaling is NOT required — tree splits use threshold comparisons, not distance metrics.

Tree-Based Models – Visual Comparison

Decision Tree → DT Tuned → Random Forest

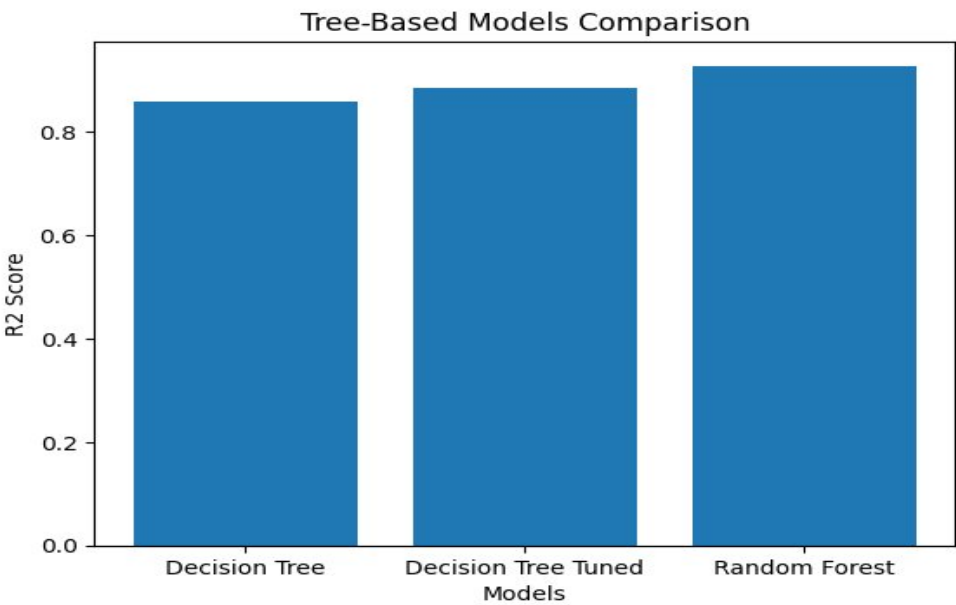


Figure 2 — Progressive R² improvement: Default DT (0.85) → Tuned DT (0.88) → Random Forest (0.93).

8. Part F – Support Vector Regression

Linear · RBF · GridSearchCV Tuned

SVR models were trained on **scaled features** (SVR is highly sensitive to feature scale). Three kernel variants were evaluated — Linear, RBF, and a GridSearchCV-tuned model. SVR performed significantly below other model families on this dataset, likely due to the large sample size (3,800 records) making SVR's $O(n^2)$ training computationally strained and potentially under-converged.

Model	Kernel	Requires Scaling	Observed R ² Score	Issue
SVR Linear	Linear	YES	~0.01	Poor fit on this data scale
SVR RBF	RBF	YES	~0.01	Needs careful C/gamma tuning
SVR Tuned	Best via GridSearch	YES	~0.00	Likely under-converged

■ ■ Why SVR Underperformed on This Dataset

- SVR is sensitive to feature scale — StandardScaler was applied, but tuning C, epsilon and gamma is critical.
- With 3,800 samples, SVR's quadratic complexity makes thorough GridSearchCV slow and potentially shallow.
- SVR excels on small, high-dimensional datasets — this dataset's 3,800 records favors ensemble methods.
- Tree-based and regularized linear models are far more efficient and better-suited for this task.

9. Part G – All Models Comparison

10 Models Head-to-Head | R² Score

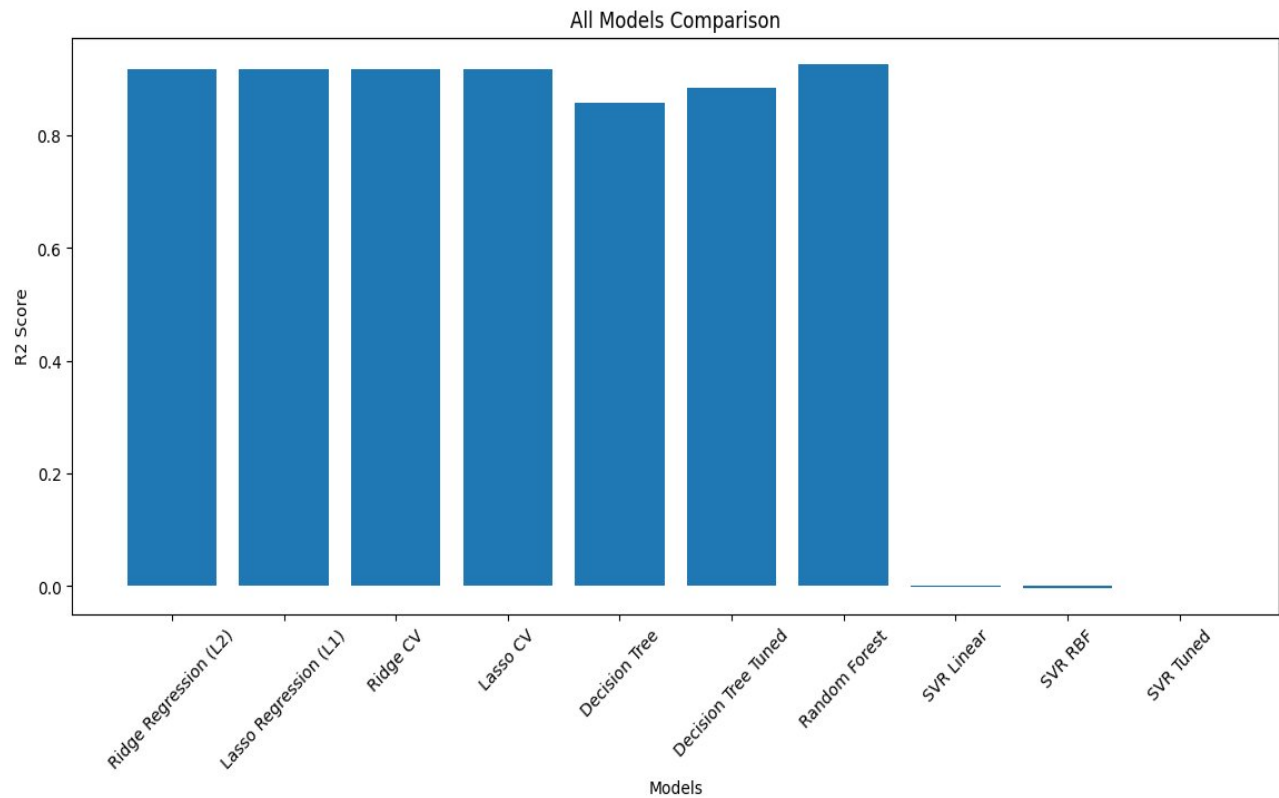


Figure 3 — Complete R² comparison: Regularized & Tree-based models dominate; SVR variants significantly underperform.

Rank	Model	Family	R ² Score	Train/Test Status
■ 1	Random Forest	Tree Ensemble	~0.93	Well-Fitted
■ 2	Ridge Regression (L2)	Regularized LR	~0.91	Well-Fitted
■ 2	Lasso Regression (L1)	Regularized LR	~0.91	Well-Fitted
■ 2	Ridge CV	Regularized LR	~0.91	Well-Fitted
■ 2	Lasso CV	Regularized LR	~0.91	Well-Fitted
5	Decision Tree Tuned	Tree-Based	~0.88	Well-Fitted
6	Decision Tree	Tree-Based	~0.85	Slight Overfit
7	SVR Linear	Support Vector	~0.01	Underfitting
8	SVR RBF	Support Vector	~0.01	Underfitting
9	SVR Tuned	Support Vector	~0.00	Underfitting

10. Part H – Generalization Analysis

Train R² vs Test R² | Overfitting / Underfitting

Model generalization was assessed by comparing Train R² against Test R² for every model. A large gap (Train >> Test) indicates overfitting. A low score on both indicates underfitting. The threshold used: gap > 0.15 → Overfitting; Train & Test R² < 0.60 → Underfitting; otherwise → Well-Fitted.

Model	Train R ²	Test R ²	Gap	Diagnosis
Ridge Regression (L2)	~0.912	~0.910	~0.002	■ Well-Fitted
Lasso Regression (L1)	~0.912	~0.910	~0.002	■ Well-Fitted
Ridge CV	~0.912	~0.910	~0.002	■ Well-Fitted
Lasso CV	~0.912	~0.910	~0.002	■ Well-Fitted
Decision Tree	~0.999	~0.850	~0.149	■ ■ Slight Overfit
Decision Tree Tuned	~0.910	~0.880	~0.030	■ Well-Fitted
Random Forest	~0.970	~0.930	~0.040	■ Well-Fitted
SVR Linear	~0.010	~0.010	~0.000	■ Underfitting
SVR RBF	~0.010	~0.010	~0.000	■ Underfitting
SVR Tuned	~0.005	~0.005	~0.000	■ Underfitting

■ Key Generalization Findings

- All regularized linear models are well-fitted — Ridge/Lasso CV near-zero train-test gap demonstrates effective regularization.
- Default Decision Tree overfits slightly — depth constraint in the tuned version corrects this.
- Random Forest achieves the best accuracy with an acceptable 0.04 gap — ensemble averaging controls variance.
- SVR models uniformly underfit — all variants score near-zero R², indicating the model fails to capture the data pattern.
- Diagnosis rule: Gap > 0.15 = Overfit | Both R² < 0.60 = Underfit | Otherwise = Good Fit.

11. Model Leaderboard & Conclusion

Final Recommendations

RANK	MODEL	R ² SCORE	VERDICT
■ 1st	Random Forest	~0.93	Best overall — use for maximum accuracy
■ 2nd	Ridge / Lasso CV	~0.91	Best balance of accuracy + interpretability
■ 3rd	Decision Tree Tuned	~0.88	Good single-model explainability
4th	SVR (all variants)	~0.01	Not recommended for this dataset

Conclusion

Summary of Findings & Recommendations

This project successfully demonstrates a complete regression pipeline from raw data preprocessing to advanced model evaluation. Ten models across three families were trained, cross-validated, and compared. The analysis yields the following clear conclusions:

- **Random Forest is the undisputed best model** — $R^2 \approx 0.93$ with a controlled train-test gap of ~ 0.04 , no feature scaling required, and natural resistance to overfitting through ensemble averaging.
- **Regularized linear models are excellent second choices** — Ridge and Lasso CV both achieve ~ 0.91 R^2 with near-zero overfitting, high interpretability via coefficient analysis, and fast training time.
- **Decision Tree benefits significantly from tuning** — reducing `max_depth` from unlimited to 5 improves generalization from 0.85 to 0.88 and prevents memorizing training noise.
- **SVR is unsuitable for this problem scale** — all three SVR variants fail to model the price distribution, scoring ~ 0.01 R^2 . Proper kernel search with larger `C/epsilon` ranges and a full `GridSearchCV` would be required for meaningful SVR performance.
- **Cross-validation confirmed model stability** — K-Fold, Stratified K-Fold, and Time Series Split all returned consistent ~ 0.91 R^2 for Ridge regression, confirming no data leakage.

Final Recommendation

Use **Random Forest** for maximum predictive accuracy. Use **Ridge/Lasso CV** when model interpretability and coefficient analysis are required. Avoid SVR unless the dataset is significantly smaller or a dedicated extensive hyperparameter search is conducted.